Project Report: The AI Research Assistant

A Full-Stack Web Application for Advanced Document Summarization and Interrogation

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1.0 Abstract

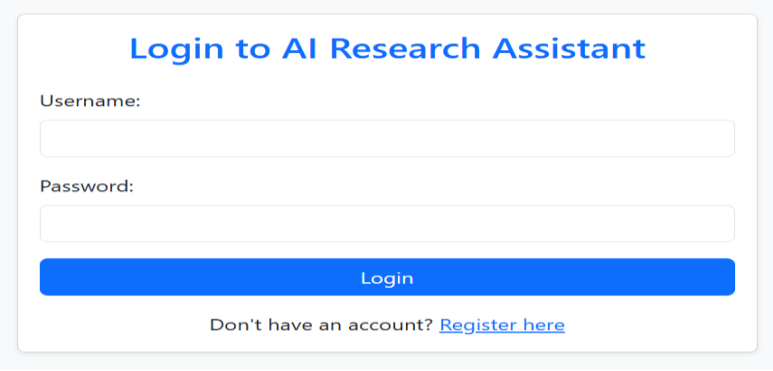
The AI Research Assistant is a full-stack web application designed to streamline the research process by leveraging advanced AI capabilities. Built with the **Flask** web framework and powered by the **Google Gemini API** via **LangChain**, the platform provides authenticated users with a suite of tools for information synthesis. Key features include a multi-source summarizer capable of processing topics, URLs, and PDF documents, and a powerful **Retrieval-Augmented Generation (RAG)** system that allows users to have interactive, in-depth conversations with the content of their uploaded documents. All user activity is securely stored in a **SQLite** database, providing a persistent history of their research.

2.0 System Architecture and Core Features

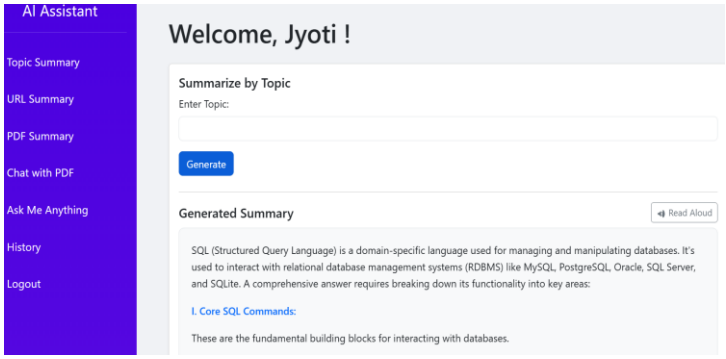
The application is designed as a monolithic web service with a clear separation of concerns, integrating a user-facing frontend, a robust backend server, a database, and external AI services.

2.1 Key Features

- **Secure User Authentication**: A complete user management system allows for user registration and login. Sessions are secured using **JSON Web Tokens (JWT)**, which are managed as secure browser cookies, ensuring that user data and history are protected and accessible only to the authenticated user.

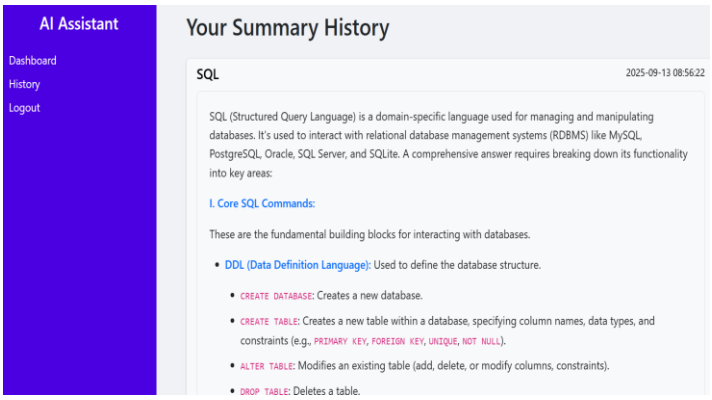


- **Multi-Source Summarization Tool**: This versatile feature enables users to generate detailed, hierarchically structured summaries from three distinct input types:
    - **Topic-Based Summary**: The user provides a topic, and the AI leverages its general knowledge and search capabilities to generate a comprehensive overview.



- **URL Summary**: The system automatically extracts the textual content from a given URL and condenses it into a detailed summary.
- **PDF Document Summary**: Users can upload a PDF file, from which the application extracts the text and produces a thorough, in-depth summary ideal for academic papers or long reports.

- **Interactive PDF Chat (RAG System)**: This is the application's core innovation. It transforms a static PDF into a conversational knowledge base. Users first "process" a PDF, which indexes its content. Afterward, they can ask specific questions and receive answers that are generated exclusively from the information contained within the document, providing accurate, context-aware responses.
- **Persistent User History**: Every summary generated by a user is automatically saved to their personal history. This feature allows users to review, revisit, and track their past research activities, creating a valuable personal knowledge repository. Each entry is timestamped and includes the original topic, the generated summary, and any associated AI-generated imagery.



3.0 Technical Implementation: The RAG Pipeline

The Interactive PDF Chat is powered by a sophisticated Retrieval-Augmented Generation (RAG) pipeline, implemented using the LangChain library. The process is divided into two distinct stages:

3.1 Stage 1: Ingestion and Indexing

This one-time process occurs when a user uploads a new PDF for processing.

1. **Text Extraction**: The PyPDFLoader utility is used to load and parse the content of the uploaded PDF file.
2. **Chunking**: The extracted text is then divided into smaller, semantically coherent chunks using the CharacterTextSplitter. This is crucial for managing context limits and improving the accuracy of the retrieval process.
3. **Embedding**: Each text chunk is converted into a high-dimensional numerical vector using **Google's embedding-001 model**. These "embeddings" capture the semantic meaning of the text.
4. **Vector Storage**: The generated embeddings are indexed and stored in a **FAISS** (Facebook AI Similarity Search) vector store. This creates a highly efficient, searchable knowledge base of the document's content, which is saved locally on the server.

3.2 Stage 2: Retrieval and Generation

This stage occurs every time a user asks a question about the processed document.

1. **Query Embedding**: The user's question is also converted into an embedding using the same model.
2. **Similarity Search**: The system performs a similarity search within the FAISS vector store to find the text chunks whose embeddings are most similar to the question's embedding. These are the most relevant parts of the document.
3. **Context-Aware Prompting**: The retrieved chunks (the "context") are inserted into a carefully engineered prompt along with the user's original question.
4. **Answer Generation**: This complete prompt is sent to the **Gemini 1.5 Flash** model, which generates a final answer based *only* on the provided context, ensuring the response is faithful to the source document.

4.0 Technology Stack

- **Backend**: Flask, Python 3
- **Database**: SQLite3
- **AI/LLM Orchestration**: LangChain
- **AI Models**: Google Gemini 1.5 Flash (Chat), Google embedding-001 (Embeddings), Hugging Face Model (Image Generation)
- **Vector Database**: FAISS (Facebook AI Similarity Search)
- **Authentication**: Flask-JWT-Extended

- **Frontend**: HTML, CSS, JavaScript (rendered via Flask templates)

5.0 Conclusion and Future Work

This project successfully demonstrates the integration of multiple modern AI technologies into a single, cohesive, and user-centric web application. It provides a practical solution for automating and enhancing the research workflow.

Potential future enhancements include:

- **Enhanced Security**: Implementing password hashing (bcrypt or Werkzeug.security) to replace plaintext password storage.
- **Asynchronous Processing**: Moving the AI generation and PDF processing tasks to a background worker queue (e.g., Celery, Redis) to improve UI responsiveness.
- **Expanded Document Support**: Adding support for other file types like .docx, .txt, and .pptx.
- **Advanced RAG Techniques**: Incorporating more sophisticated retrieval strategies, such as re-ranking or query transformations, to further improve the accuracy of the chat feature.